



南 京 大 学 作 业 纸

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习题 8.4

1(2) $xy + z \ln x + 3 \ln y - 1 = 0$

设 $F(x, y) = xy + z \ln x + 3 \ln y - 1$,

则 $F(x, y) \Rightarrow F'_x = y + \frac{z}{x}, F'_y = x + \frac{3}{y}$ 都连续

$F(1, 1) = 0$ 符合

$F'_y(1, 1) = 4 \neq 0$ 符合

$\therefore$ 存在以 x 为自变量的隐函数

$$\frac{dy}{dx} = -\frac{F'_x}{F'_y} = -\frac{y + \frac{z}{x}}{x + \frac{3}{y}} = -\frac{y(xy+z)}{x(xy+3)}$$

2(1) $x+y+z = \cos(xyz) = 0$ 在点 $(0, 0, -1)$

$F'_x = 1 + \sin(xyz) yz$ 连续

同理 $F'_y = 1 + \sin(xyz) xz$ 连续

$F(0, 0, -1) = 1 - 1 = 0$ 符合

$F'_x(0, 0, -1) = 1 + 0 \neq 0$

$F'_y(0, 0, -1) = 1 + 0 \neq 0$ 都符合

$\therefore$ 存在以 x, y 为自变量的隐函数

$$\frac{\partial z}{\partial x} = -\frac{F'_x}{F'_z} = -\frac{1 + \sin(xyz) yz}{-1 + \sin(xyz) xy} = \frac{1 + yz \sin(xyz)}{1 - xy \sin(xyz)}$$

$$\frac{\partial z}{\partial y} = -\frac{F'_y}{F'_z} = -\frac{1 + \sin(xyz) xz}{\sin(xyz) xy - 1} = \frac{1 + xz \sin(xyz)}{1 - \sin(xyz) xy}$$

3(1) $\ln \sqrt{x^2+y^2} = \arctan \frac{y}{x}$

两边对 x 求导 $\frac{1}{\sqrt{x^2+y^2}} \cdot \frac{1}{2} \frac{2x+2yy'}{\sqrt{x^2+y^2}} = \frac{x+yy'}{x^2+y^2}$

$$= \frac{1}{(\frac{y}{x})^2+1} \cdot \frac{y'x-y}{x^2} = \frac{y'x-y}{x^2+y^2}$$

$$\Rightarrow x+yy' = y'x-y \Rightarrow y' = \frac{x+y}{x-y} = \frac{dy}{dx}$$

3(3) $\cos^2 x + \cos^2 y + \cos^2 z = 1$ 求 dz

利用隐函数微分

$$\frac{\cos 2x + 1}{2} + \frac{\cos 2y + 1}{2} + \frac{\cos 2z + 1}{2} = 1$$

$$\cos 2x + \cos 2y + \cos 2z = -1$$

$$2 \sin 2x dx + 2 \sin 2y dy + 2 \sin 2z dz + 1 = 0$$

$$dz = -\frac{\sin 2x dx + \sin 2y dy}{\sin 2z}$$

$$= -\frac{\sin 2x}{\sin 2z} dx - \frac{\sin 2y}{\sin 2z} dy$$

3(5) $x+y+z = e^{-(x+y+z)}$

两边对 x 求导 $1 + \frac{\partial z}{\partial x} = e^{-(x+y+z)} \left[-1 - \frac{\partial z}{\partial x} \right]$

$$\Rightarrow \left[e^{-(x+y+z)} + 1 \right] \left(1 + \frac{\partial z}{\partial x} \right) = 0$$

$$\Rightarrow \frac{\partial z}{\partial x} = -1$$

同理对 y 求导

$$1 + \frac{\partial z}{\partial y} = e^{-(x+y+z)} \left[-1 - \frac{\partial z}{\partial y} \right]$$

$$\Rightarrow \text{同理得 } \frac{\partial z}{\partial y} = -1$$

$$12(1) \begin{cases} z = x^2 + y^2 \\ x^2 + y^2 + 3z^2 = 20 \end{cases}$$

$$\frac{dy}{dx} \Rightarrow$$

对 \$x\$ 求导

$$\begin{cases} \frac{dz}{dx} = 2x + 2y \frac{dy}{dx} \\ 2x + 4y \frac{dy}{dx} + 6z \frac{dz}{dx} = 0 \end{cases}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-x(6z+1)}{2y(3z+1)}$$

$$\Rightarrow \frac{dz}{dx} = \frac{x}{3z+1}$$

习题 8.6

2(2) \$u = x^2 - xy + z^2\$ 在 \$(1, 1, 1)\$ 处, 从 \$(1, 0, 1)\$ 到 \$(3, -1, 3)\$ 的弧

$$u'_x = 2x - y, \quad u'_y = -x, \quad u'_z = 2z.$$

$$\text{在 } x=y=z=1 \text{ 处 } u'_x=2, \quad u'_y=-1, \quad u'_z=2$$

$$\vec{r} = (3, -1, 3) - (1, 0, 1) = (2, -1, 2) \quad |\vec{r}| = \sqrt{2^2 + (-1)^2 + 2^2} = 3$$

$$\vec{r} = \left(\frac{2}{3}, -\frac{1}{3}, \frac{2}{3} \right)$$

$$\therefore \frac{\partial f}{\partial l} = \frac{\partial f}{\partial x} \cos \alpha + \frac{\partial f}{\partial y} \cos \beta + \frac{\partial f}{\partial z} \cos \gamma$$

$$= \frac{2}{3} \times 2 + \left(-\frac{1}{3}\right) \times (-1) + \frac{2}{3} \times 2 = 3.$$

\$\therefore\$ 切线为 3

$$12(3) \begin{cases} x = e^u + u \sinh v \\ y = e^u - u \cosh v \end{cases}$$

$$\text{求 } \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$$

对 \$x\$ 求偏导得

$$\begin{cases} 1 = e^u \frac{\partial u}{\partial x} + \frac{\partial u}{\partial x} \sinh v + u \cosh v \frac{\partial v}{\partial x} \\ 0 = e^u \frac{\partial u}{\partial x} - \frac{\partial u}{\partial x} \cosh v + u \sinh v \frac{\partial v}{\partial x} \end{cases}$$

$$\Rightarrow \begin{cases} (e^u + \sinh v) \frac{\partial u}{\partial x} + u \cosh v \frac{\partial v}{\partial x} = 1 \\ (e^u - \cosh v) \frac{\partial u}{\partial x} + u \sinh v \frac{\partial v}{\partial x} = 0 \end{cases}$$

$$\Rightarrow \frac{\partial u}{\partial x} = - \frac{\begin{vmatrix} 1 & u \cosh v \\ 0 & u \sinh v \end{vmatrix}}{\begin{vmatrix} e^u + \sinh v & u \cosh v \\ e^u - \cosh v & u \sinh v \end{vmatrix}} = \frac{-u \sinh v}{u e^u \sinh v + u \sinh^2 v - u \cosh v e^u + u \cosh^2 v}$$

$$= \frac{e^u (\sinh v - \cosh v) + 1}{e^u (\sinh v - \cosh v) + 1}$$

$$\text{同理 } \frac{\partial u}{\partial y} = \frac{-\cosh v}{e^u (\sinh v - \cosh v) + 1}, \quad \frac{\partial v}{\partial x} = \frac{\cosh v - e^u}{u [e^u (\sinh v - \cosh v) + 1]}$$

$$\frac{\partial v}{\partial y} = \frac{\sinh v + e^u}{u [e^u (\sinh v - \cosh v) + 1]}$$

12(1) \$u = 2x^3y - 3y^2z\$ 在 \$(1, 2, -1)\$ 处

$$\frac{\partial f}{\partial l} = \frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j} + \frac{\partial f}{\partial z} \vec{k}$$

$$\left. \frac{\partial f}{\partial x} \right|_{(1, 2, -1)} = 6x^2y \Big|_{(1, 2, -1)} = 12$$

$$\left. \frac{\partial f}{\partial y} \right|_{(1, 2, -1)} = 2x^3 - 6yz \Big|_{(1, 2, -1)} = 14$$

$$\left. \frac{\partial f}{\partial z} \right|_{(1, 2, -1)} = -3y^2 = -12$$

$$\therefore \frac{\partial f}{\partial l} = 12 \vec{i} + 14 \vec{j} - 12 \vec{k}$$

切线为 \$12\vec{i} + 14\vec{j} - 12\vec{k}\$



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习题 8.7

2. $x=1, y=2t, z=3t^2$

方向向量 $(1, 2t, 3t^2)$

$x+2y+z=4$ 法向量为 $(1, 2, 1)$

$(1, 2t, 3t^2) \perp (1, 2, 1)$

$1+4t+3t^2=0$

$\Rightarrow t=-1$ 或 $-\frac{1}{3}$

当 $t=-1$ 时, 切点 $(-1, -2, 3)$

$t=-\frac{1}{3}$ 时, 切点 $(-\frac{1}{3}, -\frac{2}{3}, \frac{1}{3})$

4. $F(x, y, z) = x^2 + y^2 + z^2 - 21$

$\frac{\partial F}{\partial x} = 2x, \frac{\partial F}{\partial y} = 2y, \frac{\partial F}{\partial z} = 2z$

在 (x_0, y_0, z_0) 处梯度是法向量

$(2x_0, 2y_0, 2z_0)$

$x+4y+6z=0$ 法向量为 $(1, 4, 6)$

$\Rightarrow \frac{x_0}{1} = \frac{y_0}{4} = \frac{z_0}{6} = k$

$(x_0, y_0, z_0) = (k, 4k, 6k)$

在 $F(x, y, z)=0$ 上

$\Rightarrow k^2 + 2(2k)^2 + 3(2k)^2 = 21$

$k = \pm 1$

有两个切平面

$k=1$, 切点 $(1, 4, 6)$

$k=-1$, 切点 $(-1, -4, -6)$

$\Rightarrow x+4y+6z = \pm 21$

8(1) $z = x^2 + 3xy^2 - 15x - 12y$

$\frac{dz}{dx} = 2x + 3y^2 - 15 = 0$

$\frac{dz}{dy} = 6xy - 12 = 0$

$\Rightarrow \begin{cases} x=1 \\ y=2 \\ z=-26 \end{cases} \quad \begin{cases} x=2 \\ y=1 \\ z=-28 \end{cases} \quad \begin{cases} x=-1 \\ y=-2 \\ z=26 \end{cases} \quad \begin{cases} x=-2 \\ y=-1 \\ z=28 \end{cases}$

在驻点处有四个

$z_{\min} = -28, z_{\max} = 28$

极大值点为 $(2, 1)$, 极大值为 -28

极大值点为 $(-2, -1)$, 极大值为 28

14(1)

$z = x^2 + y^2, \frac{x}{a} + \frac{y}{b} = 1$

$\frac{x}{a} + \frac{y}{b} = 1$

$\Rightarrow x = (1 - \frac{y}{b})a$

$z = a^2(1 - \frac{y}{b})^2 + y^2$

$= (1 - \frac{a^2}{b^2} + 1)y^2 - \frac{2a^2}{b}y + a^2$

$= (\frac{a^2+b^2}{b^2}) [y^2 - \frac{2a^2b}{a^2+b^2}y + \frac{a^2b^2}{a^2+b^2}]$

对 y 求导 $y = \frac{a^2b}{a^2+b^2}$, 极大值为 $\frac{a^2b^2}{a^2+b^2}$

17. $f(x, y) = x^3 - y^2 + 2$ 在 $D = \{(x, y) \mid x^2 + \frac{y^2}{4} \leq 1\}$ 上求极值.

$$\begin{cases} f'_x(x, y) = 3x^2 \\ f'_y(x, y) = -2y \end{cases}, \text{ 考虑在 } x^2 + \frac{y^2}{4} = 1 \text{ 上求极值}$$

$$L(x, y) = x^3 - y^2 + 2 - \lambda(x^2 + \frac{y^2}{4} - 1)$$

$$\begin{cases} L'_x(x, y) = 3x^2 - 2\lambda x = 0 \\ L'_y(x, y) = -2y - \frac{\lambda}{2}y = 0 \\ x^2 + \frac{y^2}{4} = 1 \end{cases}$$

$$\lambda = \frac{3x}{2} = \frac{3}{2}x, \quad x = \frac{2}{3}\lambda$$

当

$y = 0$ 时, $x = \pm 1$, 极值点为 $f(1, 0) = 3$

当 $x = 0$ 时, $y = \pm 2$, 极值点为 $f(0, 2) = -2$

$\therefore$ 最大值为 3, 最小值为 -2.

19. $z = x^2 + y^2$, $x + y - 2z = 2$ 之间的最短距离.

(x, y) , $x^2 + y^2$ 在 $x + y - 2z = 2$ 上求极值

$$d = \frac{|x + y - 2(x^2 + y^2) - 2|}{\sqrt{1^2 + 1^2 + 4}}$$

$$= \frac{2|(x - \frac{1}{4})^2 + (y - \frac{1}{4})^2 + \frac{7}{8}|}{\sqrt{6}}$$

$$\geq \frac{2 \times \frac{7}{8}}{\sqrt{6}} = \frac{7}{4\sqrt{6}}$$

当且仅当 $x = y = \frac{1}{4}$ 时, $d_{\min} = \frac{7}{4\sqrt{6}}$